

SURAT THANI, Thailand

WMO: 485510

Lat: 9.1335N

Lon: 99.1521E

Elev: 7

StdP: 101.24

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.6	21.4	18.5	13.4	26.5	19.4	14.2	26.1	8.5	28.1	7.9	28.1	0.2	330	0.303

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.4	35.8	26.1	35.0	26.2	34.2	26.2	28.1	32.1	27.6	31.8	27.2	31.6	3.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.8	26.7	22.3	29.6	26.1	21.6	29.4	90.2	31.5	87.9	32.4	86.2	31.8	33.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.5	5.7	18.6	37.4	1.3	1.3	17.7	38.3	16.9	39.1	16.2	39.8	15.2	40.8		
			17.5	29.7	1.6	1.6	16.4	30.9	15.4	31.8	14.5	32.7	13.4	33.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	26.3	27.0	28.2	29.2	29.1	28.5	28.2	28.1	27.9	27.5	26.8	26.2
	DBStd	1.44	1.04	1.01	1.06	1.12	1.18	1.09	0.98	1.04	0.99	1.01	1.07	1.15
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6482	506	475	566	577	591	556	564	561	538	543	504	502
	CDD18.3	3441	247	242	307	327	333	306	306	303	288	284	254	243
	CDH23.3	32911	2118	2404	3346	3715	3440	3079	2921	2835	2555	2419	2125	1953
	CDH26.7	13114	714	1026	1603	1819	1488	1267	1162	1126	957	806	615	532
(14) Wind	WSAvg	1.8	2.0	2.0	2.0	1.6	1.5	1.8	1.9	1.9	1.6	1.3	1.6	2.0
(15) Precipitation	PrecAvg	1832	53	24	78	81	181	161	179	173	215	251	280	158
	PrecMax	2335	283	161	769	228	257	255	279	458	369	637	597	549
	PrecMin	1368	1	0	0	3	82	80	78	74	94	109	82	47
	PrecStd	246	59	36	146	62	50	52	57	73	74	113	147	120
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	34.2	36.1	37.4	37.1	35.2	35.0	35.0	34.4	33.9	33.0	32.2
		MCWB	25.3	25.0	25.1	26.1	26.3	26.4	26.1	26.1	26.5	26.3	26.3	26.0
	2%	DB	31.9	33.1	35.0	36.0	35.2	34.2	33.9	34.0	33.6	32.7	31.9	31.2
		MCWB	25.1	25.1	25.5	26.0	26.8	26.3	26.1	26.1	26.4	26.3	26.2	25.8
	5%	DB	31.1	32.2	34.0	35.0	34.1	33.4	33.0	33.0	32.7	31.8	31.0	30.4
		MCWB	25.0	25.1	25.5	26.3	26.8	26.3	26.0	26.0	26.1	26.2	26.2	25.5
	10%	DB	30.2	31.7	33.1	33.9	33.0	32.5	32.0	32.0	31.5	30.8	30.0	29.6
		MCWB	24.8	25.0	25.5	26.2	26.7	26.3	25.9	25.9	26.0	26.1	25.8	25.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	27.0	27.7	28.2	28.6	28.4	27.8	27.7	27.9	28.0	28.0	27.6
		MCDB	30.5	31.3	32.3	33.2	33.1	32.2	31.3	31.8	31.2	31.0	30.1	29.9
	2%	WB	26.2	26.2	26.9	27.5	27.9	27.6	27.2	27.1	27.1	27.1	27.0	26.7
		MCDB	29.9	30.9	32.1	32.8	32.5	31.8	31.3	31.5	31.4	30.8	30.3	29.7
	5%	WB	25.8	25.9	26.5	27.1	27.4	27.1	26.7	26.7	26.7	26.7	26.5	26.2
		MCDB	29.5	30.6	31.7	32.6	32.1	31.4	31.0	31.0	30.9	30.3	29.7	29.1
	10%	WB	25.3	25.5	26.1	26.7	27.0	26.6	26.3	26.3	26.2	26.2	26.1	25.7
		MCDB	28.9	30.2	31.3	32.1	31.6	30.9	30.5	30.5	30.3	29.7	29.1	28.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	10.0	10.6	10.4	9.2	8.6	8.5	8.5	8.3	7.5	6.5	6.7
		MCDBR	9.6	10.9	11.9	11.8	10.5	10.0	9.8	10.1	9.7	8.9	8.3	8.4
		MCWBR	4.1	4.3	4.1	3.7	3.7	3.6	3.5	3.6	3.7	3.6	3.7	3.9
	5% WB	MCDBR	8.4	10.2	11.0	10.8	9.8	9.1	8.9	9.0	9.0	8.2	7.5	7.3
		MCWBR	4.0	4.4	4.1	3.9	3.9	3.8	3.7	3.7	3.8	3.7	3.7	3.8
(40) Clear-Sky Solar Irradiance	taub		0.481	0.491	0.498	0.475	0.447	0.444	0.445	0.437	0.427	0.446	0.471	0.481
	taud		2.197	2.161	2.177	2.247	2.368	2.411	2.392	2.404	2.424	2.357	2.273	2.214
	Ebn at Noon		831	836	833	842	849	841	844	863	882	866	835	822
	Edh at Noon		146	156	155	143	124	117	120	121	119	126	135	141
(44) All-Sky Solar Radiation	RadAvg		4.68	5.61	5.82	5.75	4.96	4.73	4.62	4.73	4.71	4.35	4.05	4.01
	RadStd		0.42	0.36	0.50	0.34	0.33	0.27	0.31	0.31	0.27	0.40	0.37	0.40

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		+0.25	+0.39	+0.60	+0.79	+0.57	+0.66	N/A	N/A	+122	+122	
(47) Regional (15 neighbors)		+0.19	+0.35	N/A	N/A	+0.20	+0.23	N/A	N/A	N/A	N/A	

Nomenclature: See separate page